

CINDER v1.1.2

Solver Convergence Study

Final Closeout & Restart Context

Formal verification status: FROZEN

Prepared 12 September 2026 | CINDER release 1.1.2 | LSODA hybrid time integration

Closeout decision. No further full convergence sweep is required for the v1.1.2 solver-verification claim. The 40-point formal dataset is frozen. The 4,945-point overnight sweep is retained as exploratory confirmation and as evidence about breakdown boundaries, hybrid non-monotonicity, gross outcome robustness, and high-throughput solver settings.

Executive snapshot

Item	Final result
Tight numerical reference	rtol = 1e-7, atol = 1e-10, max step = 2.5 ms
Research / verification setting	rtol = 1e-4, atol = 1e-7, max step = 10 ms
Production candidate	rtol = 1e-3, atol = 1e-6, max step = 50 ms
Formal sweep	40 points; 39 completed; 25 passed all review guards
Dense exploratory sweep	4,945 points; 4,681 completed; 4,072 exact reference signatures
Formal convergence region	All tested max steps pass for rtol <= 1e-4
Main numerical insight	Mechanics-level convergence is stricter than macroscopic outcome convergence

1. What this study was trying to establish

The question was not merely whether LSODA reaches the final time. The study asks whether CINDER's time trajectory is a property of the modeled mechanics rather than of arbitrary numerical controls. Because CINDER is hybrid, that requires checking both the continuous state evolution and the discrete contact/regime history.

- Continuous-state convergence: primary speed, secondary speed, belt speed, shift position, and shift speed.
- Hybrid convergence: exact transition sequence, transition timing, and duration spent in different regimes.
- Tolerance sensitivity: relative tolerance, maximum step, and an independent absolute-tolerance sweep.
- Computational cost: integration wall time and native solve_ivp point count as transparent cost proxies.

2. Final comparison method

Important methodological fix. Raw absolute-time state differences are not a valid primary norm across reset discontinuities: an event shifted by microseconds can compare a pre-reset state against a post-reset state and create an $O(1)$ pointwise error even when the hybrid solutions have converged.

The final revision-4 metric therefore compares corresponding hybrid segments after parameterizing each segment by normalized phase. Candidate and reference states are evaluated directly from their LSODA dense interpolants at the same 2,049 phase points. Event timing is reported separately rather than being hidden by the alignment. Regime-history disagreement is computed as exact time duration over the union of candidate and reference segment boundaries, not by sampling a coarse time grid.

A second post-processing floor was also removed during development: earlier event-aligned traces had been stored at different absolute-time spacings (2 ms candidates versus 1 ms reference) and subsequently re-interpolated. Direct common-phase evaluation from each dense solution eliminated that sampling floor.

Frozen review guard	Threshold
Event-aligned normalized trajectory RMS	$\leq 1e-4$
Event-aligned normalized max error	$\leq 1e-3$
Maximum corresponding event-time error	≤ 1 ms
Exact regime-history mismatch fraction	$\leq 1e-3$
Hybrid transition signature	Exact match required

3. Frozen formal sweep: main findings

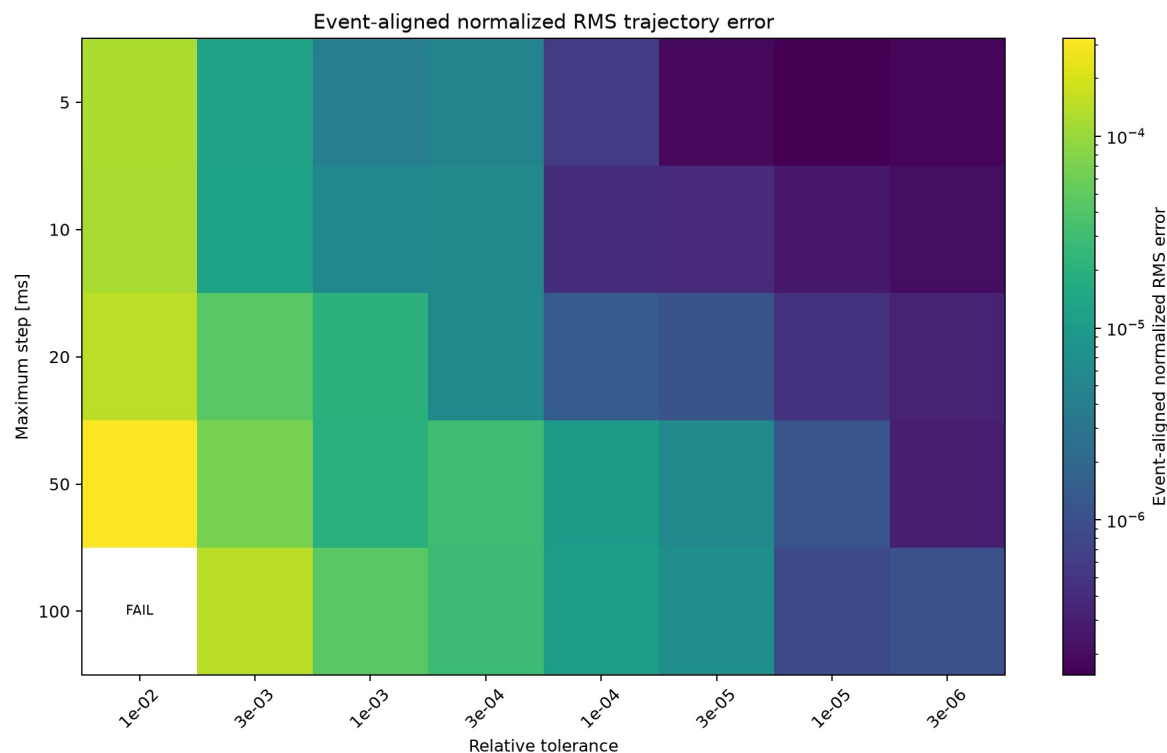


Figure 1. Formal 8 x 5 convergence grid. Event-aligned RMS error decreases by orders of magnitude toward tighter controls; the canonical point lies well inside the low-error region.

The formal sweep covered eight relative tolerances from 1e-2 to 3e-6 and five maximum steps from 100 ms to 5 ms. Thirty-nine of forty integrations completed. The only failure was the deliberately extreme $\text{rtol} = 1\text{e-}2$, max step = 100 ms case, which entered an inadmissible unilateral primary flyweight-contact state. This is treated as a numerical/model-domain failure, not evidence that the corresponding physical trajectory truly lifts off.

Robust plateau. Every tested maximum step from 5 to 100 ms passed all formal guards when $\text{rtol} \leq 1\text{e-}4$. This contiguous 20-point block is the cleanest formal convergence statement; isolated pass/fail stripes at looser tolerances are not used as a sharp convergence boundary.

Canonical setting

Metric	Canonical result
Settings	$\text{rtol } 1\text{e-}4$ $\text{atol } 1\text{e-}7$ max step 10 ms
Event-aligned trajectory RMS	4.274e-07
Event-aligned maximum error	6.962e-05
Maximum event-time error	7.99 microseconds
Exact regime mismatch	8.457e-06
Transition signature	Exact (16 transitions)
Native solver points	1156

4. Absolute tolerance and the small-state problem

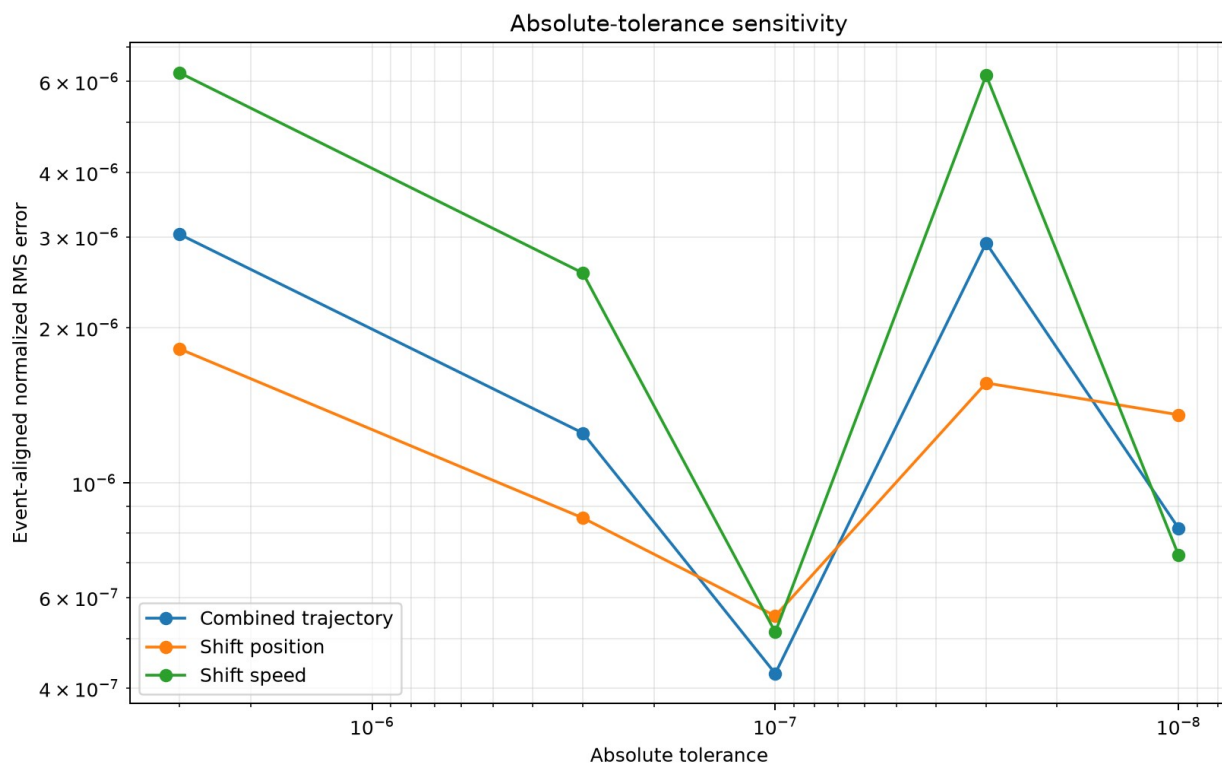


Figure 2. Independent absolute-tolerance sweep at $rtol = 1e-4$ and $max\ step = 10\ ms$. Hybrid topology is sensitive to $atol$ in the loose range and becomes stable in the tighter range.

The independent absolute-tolerance sweep showed why a scalar $atol$ cannot be chosen casually for this mixed-scale hybrid state vector. The event count was 12 at $atol = 1e-5$, 16 at $3e-6$, 14 at $1e-6$, then 16 throughout $3e-7$ to $1e-8$. The tighter range therefore recovers the reference topology, although pointwise max-error guards remain mildly non-monotonic because adaptive hybrid integration is not required to approach the reference monotonically for every local norm.

Interpretation. $atol = 1e-7$ is retained for the research setting because it lies inside the stabilized hybrid region and comfortably passes all review guards. The sweep provides empirical justification for this value rather than treating it as an arbitrary small number.

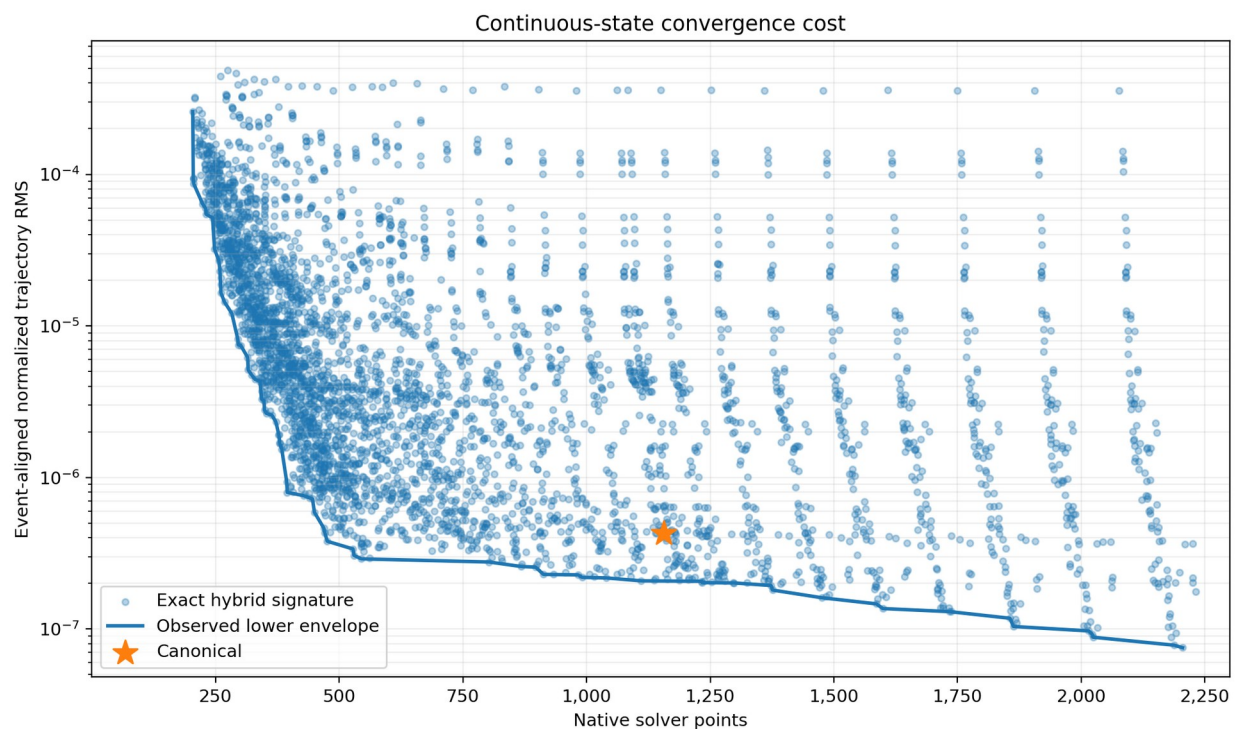


Figure 3. Dense event-aligned trajectory error versus native solver points. The lower envelope makes the diminishing-return tradeoff visible without the multiprocessing contention present in wall-time measurements.

5. Dense overnight sweep: what it added scientifically

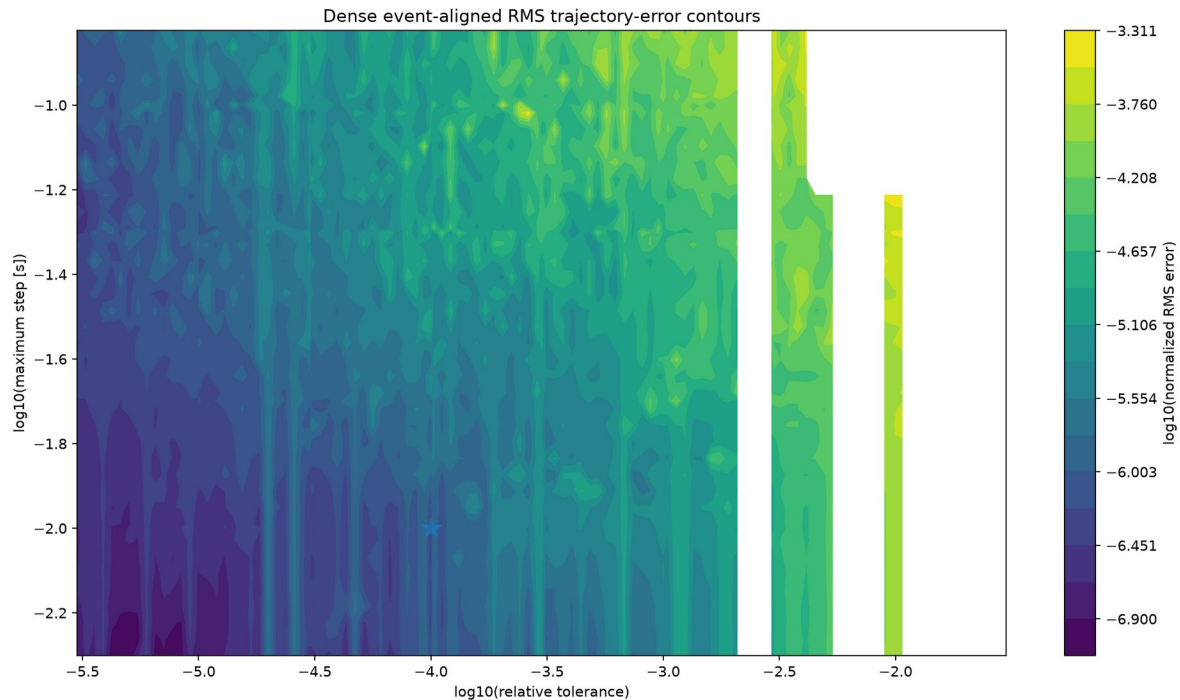


Figure 4. High-resolution exploratory error contours from the 115 x 43 dense grid. White bands are intentionally undefined event-aligned errors where the discrete transition signature differs from the reference.

Dense sweep statistic	Result
Total configurations	4,945
Completed	4,681
Failed	264
Completed with exact 16-event signature	4,072
Domain	rtol 3e-6 to 3e-2; max step 5 to 150 ms
Purpose	Exploratory; does not replace the frozen formal verification grid

The dense surface strongly confirmed the formal pattern, but it also revealed structure that forty points could only hint at. In particular, hybrid topology changes non-monotonically with tolerance: entire tolerance bands can produce 14 transitions, the correct 16-transition sequence can reappear at a looser tolerance, and still coarser bands produce 12 or 10 transitions. This is direct evidence that one scalar error norm or one monotonic refinement path is insufficient for a hybrid model.

6. Breakdown boundary and hybrid non-monotonicity

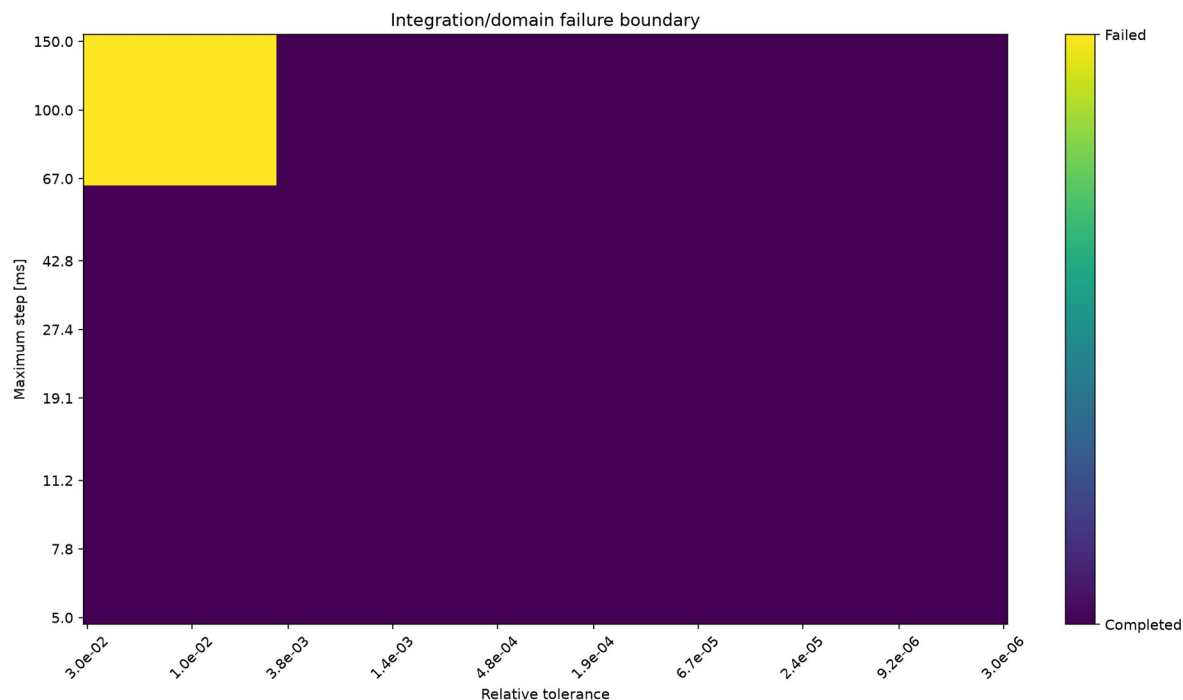


Figure 5. Dense numerical/model-domain failure boundary. All 264 failures occupy one sharply resolved coarse-control rectangle.

The dense sweep resolved a striking two-parameter failure boundary. Every configuration with $\text{rtol} \geq 0.004515$ and $\text{max step} \geq 67.0$ ms failed, and all 264 points in that rectangle failed; no completed point exists inside it. All failures are the same class: the primary fixed-pivot flyweight contact becomes inadmissible in the deadzone. Neither loose rtol alone nor large max step alone is sufficient - their combination triggers the breakdown.

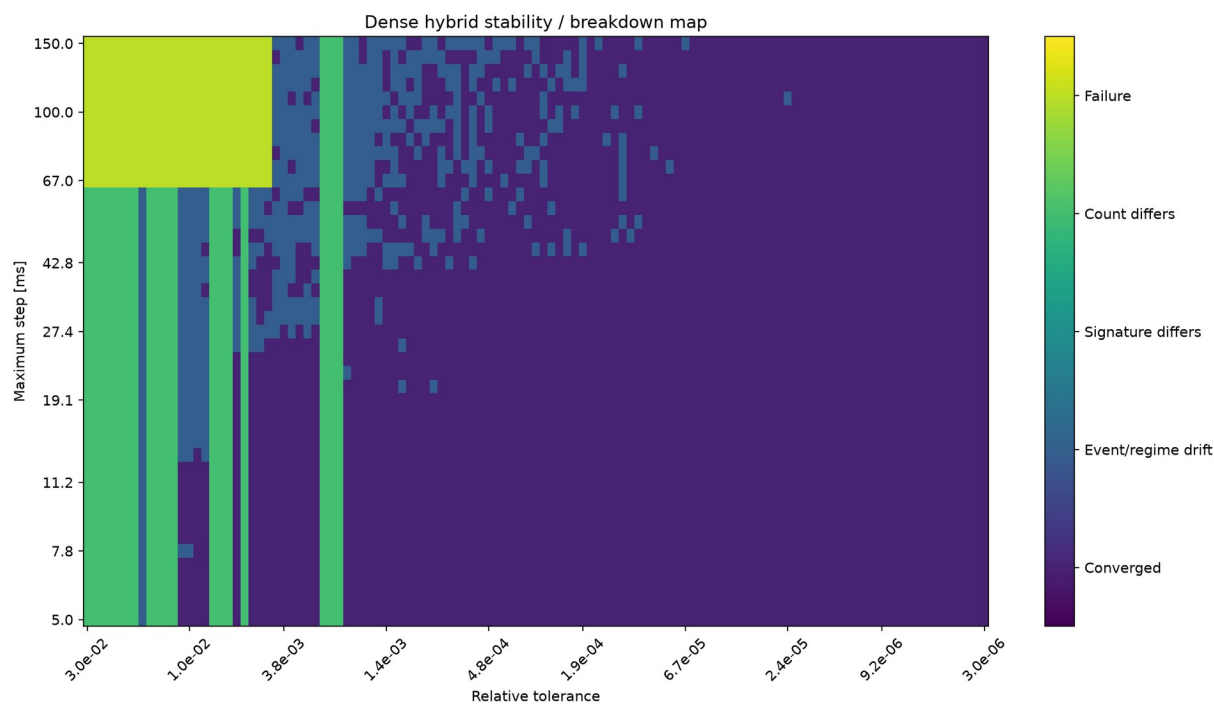


Figure 6. Dense hybrid-stability map. The discrete transition sequence disappears and reappears in tolerance bands before eventual numerical/domain failure.

One especially clear band occurs from approximately $\text{rtol} = 2.27\text{e-}3$ to $2.69\text{e-}3$: all 43 tested max-step values complete with 14 rather than 16 transitions. At $\text{rtol} = 2.94\text{e-}3$ the 16-event reference signature returns across the whole max-step axis. Still coarser successful bands alternate among 16, 14, 12, and 10 events. This non-monotonic behavior is expected to be possible in an adaptive hybrid solver and reinforces the decision to audit topology explicitly.

7. Mechanics-level convergence versus outcome-level robustness

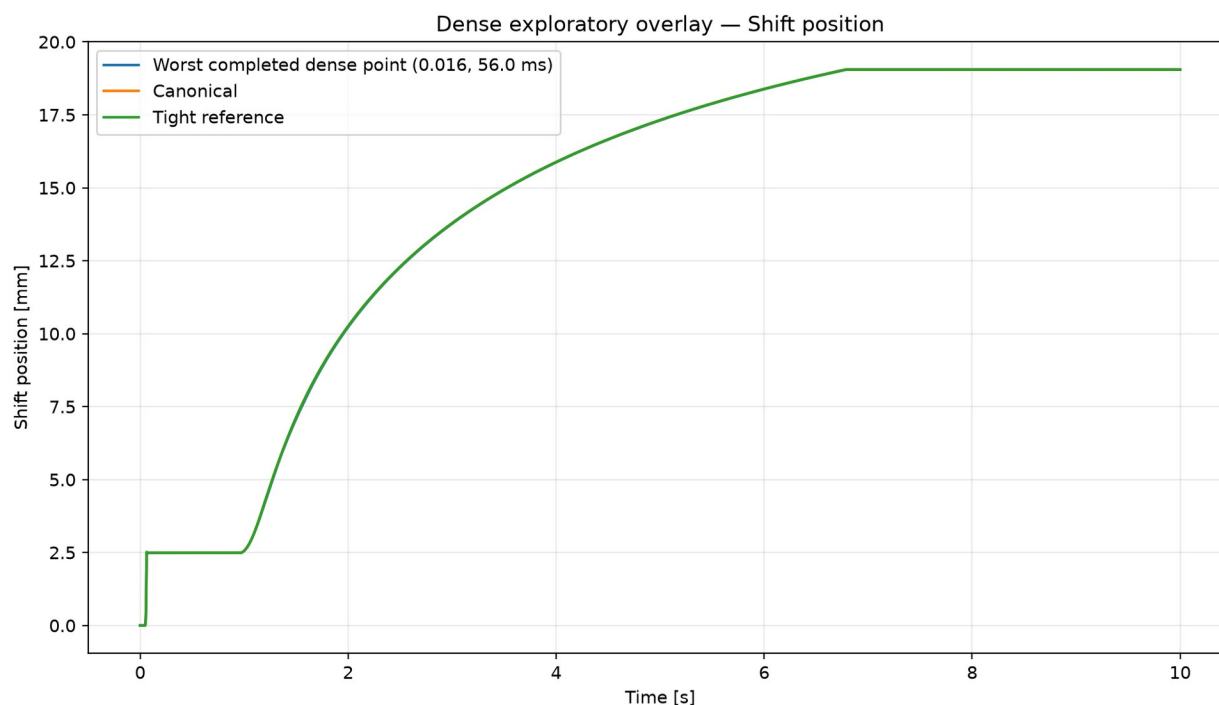


Figure 7. Tight reference, canonical solution, and the worst completed dense point that still retained the exact transition signature. Even this intentionally under-resolved run has a nearly indistinguishable macroscopic shift trajectory.

The dense sweep exposed a useful distinction. Detailed hybrid mechanics can stop agreeing before the gross launch trajectory materially changes. Among the 609 completed simulations with a different discrete signature, the median gross state-trajectory RMS (primary speed, secondary speed, belt speed, and shift position in absolute time; shift speed excluded) was only $1.225\text{e-}04$; the 90th percentile was $3.493\text{e-}04$, and the worst was $9.408\text{e-}04$. The median normalized final-state error was $2.559\text{e-}06$.

Scientific interpretation. A candidate can be locally/mechanistically wrong - for example by missing a brief stick/slip excursion - yet remain globally almost indistinguishable in RPM, belt speed, and shift position. Therefore mechanics-level convergence and outcome-level robustness should be reported as complementary claims, not substituted for one another.

For verification, the strict mechanics-level checks remain necessary because CINDER is intended to predict internal traction state, forces, slip, and transient mechanisms. For an engineering application concerned

mainly with vehicle-level outcomes, gross metrics can be a useful second layer because they better reflect practical sensitivity.

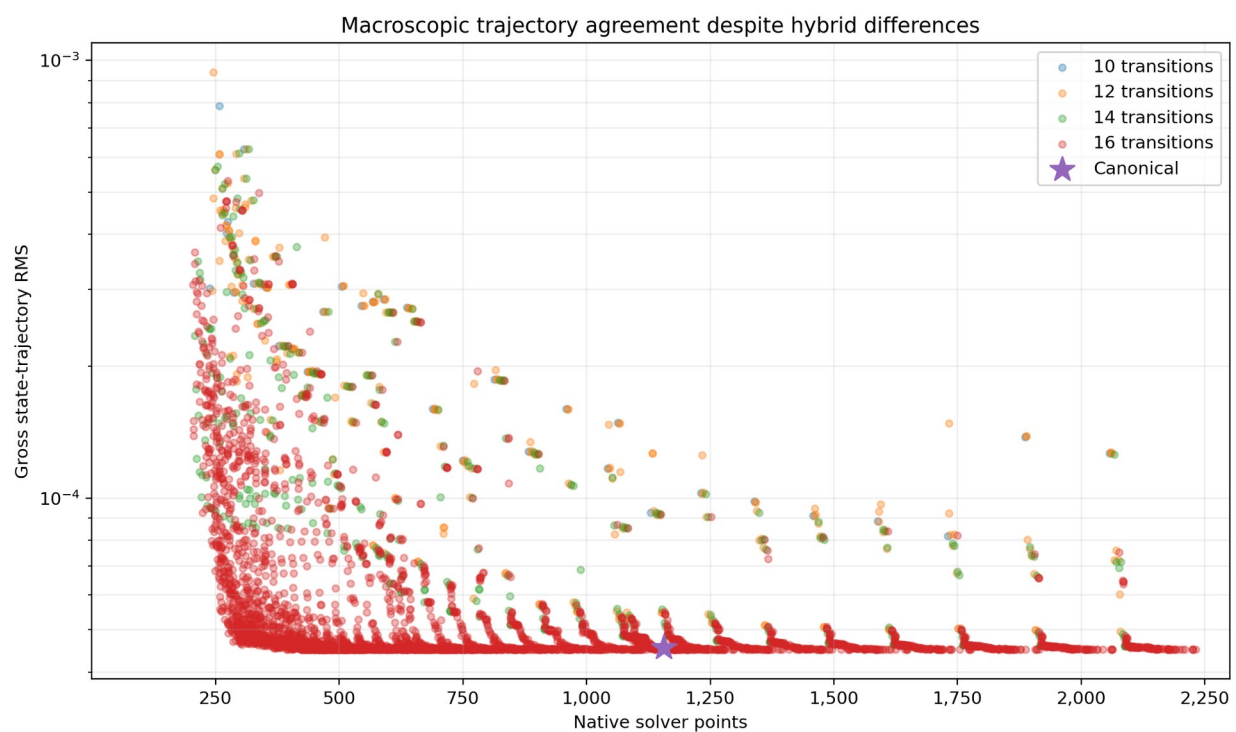


Figure 8. Gross trajectory error versus native solver points, grouped by transition count. Different hybrid signatures can remain macroscopically close to the reference.

8. Practical solver settings after the study

The convergence study now supports two different numerical settings for two different jobs. The research setting is deliberately conservative and is the one to retain for verification-quality studies. The production candidate trades a small amount of mechanics-level accuracy for substantially lower integration work and is attractive for a hosted application running many simulations.

Metric	Research / verification	High-throughput candidate
rtol	1e-4	1e-3
atol	1e-7	1e-6
Maximum step	10 ms	50 ms
Exact 16-event signature	Yes	Yes
Trajectory RMS	4.274e-07	1.973e-05
Maximum event-time error	0.008 ms	0.422 ms
Regime mismatch	8.457e-06	9.705e-05
Native solver points	1156	341
Integration wall time in dense run	9.73 s	3.35 s

Production candidate. On this benchmark, rtol = 1e-3, atol = 1e-6, max step = 50 ms was about 2.9x faster by the recorded pure-integration timer and used 70.5% fewer native solver points than the canonical setting, while retaining the exact hybrid signature and sub-millisecond event timing error.

This is a candidate application default, not yet a universal CINDER guarantee. Before using it as the permanent setting for a public/high-volume service, repeat a small comparison across several representative trajectories (for example launch, load-induced backshift, slip-rich operation, and deceleration). There is no need to repeat a 5,000-point sweep for each case; a few candidate/tight comparisons per case are sufficient.

9. Freeze decision and restart checklist

FROZEN. The formal solver-convergence study is closed for CINDER v1.1.2. Do not rerun or retune the 40-point formal dataset unless the governing equations, hybrid topology, state definition, or integration implementation changes materially.

Artifacts to preserve

- Formal study: `studies/solver-convergence/run.py`, `study.json`, and `artifacts/`.
- Publication-only companions: `publication_plots.py`.
- Exploratory dense study: `overnight_dense.py` and `artifacts/dense-overnight/`.
- Dense sweep is supplementary: it confirms and extends the formal conclusions but does not replace the pre-defined formal grid.
- The overnight explorer checkpoints `dense_sweep.csv` every 20 completed points and resumes an existing grid unless `--fresh` is requested.

If this study is reopened later

- First ask whether the model changed enough that numerical convergence must be re-established. If not, reuse this study.
- If the model changed, retain the final methodology: event-aligned direct dense-output comparison + independent event timing + exact regime-duration mismatch.
- Do not return to raw absolute-time max/RMS as the primary hybrid convergence criterion.
- For production optimization, validate the $1e-3$ / $1e-6$ / 50 ms candidate across several qualitatively distinct operating trajectories.
- Optional future efficiency work: use per-state/vector absolute tolerances based on meaningful physical resolutions. This could improve the speed/accuracy frontier because the five states have very different scales.
- Optional numerical reference check: compare the existing tight reference against one super-tight run. Useful as an extra publication-strength check, but not required to close the present sweep.

Final conclusion

The solver-convergence program supports the central numerical claim needed for CINDER v1.1.2: once the numerical controls enter a broad stable region, the hybrid transition structure, event timing, regime history, and event-aligned continuous trajectory are insensitive to further tightening. The canonical research settings lie comfortably inside this region. The dense extension further shows that macroscopic CVT response remains remarkably robust even beyond the range where detailed hybrid mechanics are identical, while a sharply resolved coarse-control boundary eventually drives the solver outside the modeled unilateral-contact domain. Together these results justify both a conservative verification setting and a substantially cheaper high-throughput candidate for future applications.

Appendix A. Commands and output locations

Formal full study: `python .\studies\solver-convergence\run.py`

Publication plots only: `python .\studies\solver-convergence\publication_plots.py`

Dense explorer plan: `python .\studies\solver-convergence\overnight_dense.py --plan-only`

Dense explorer: `python .\studies\solver-convergence\overnight_dense.py --hours 12 --workers 6`

Resume after interruption: Rerun the same `overnight_dense.py` command; do not use `--fresh`.

Dense explorer final plot set

- 01-06: accuracy, event-time, regime, and hybrid high-resolution maps/contours.
- 07: wall-time convergence dashboard.
- 08: failure boundary.
- 09-11: worst same-signature trajectory overlays.
- 12-15: native-solver-point cost plots for trajectory RMS, event timing, regime mismatch, and gross trajectory RMS.
- `dense_cost_metrics.csv`: derived cost table including the gross/macroscopic trajectory metric.